

Networks

Paths and connected components

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Optimization: principles and algorithms



Paths and connected components

There are many paths in a network.

Some are long, some are short

Later on, we will be interested in finding the shortest or the longest path between two nodes

And, sometimes, there is no path connecting two nodes.

We formalize these concepts in this video.

Longest simple path

Lemma 21.5

- ▶ Consider a directed graph with m nodes.
- ▶ The maximum number of arcs in a simple path is $m - 1$.

Longest simple path

Lemma 21.5

- ▶ Consider a directed graph with m nodes.
- ▶ The maximum number of arcs in a simple path is $m - 1$.

Proof

- ▶ Suppose that a simple path visiting all the m nodes exist.
- ▶ It has exactly $m - 1$ arcs.
- ▶ Extend by one more arc: not simple anymore.
- ▶ If there is no such path: the longest has less than $m - 1$ arcs.

Proof

Draw a small example with 5 nodes, and show the 6 permutations.

Note that each permutation does not necessarily correspond to a valid path, so the number of permutation is an upper bound on the number of paths.

Finite number of simple paths

Lemma 21.6

- ▶ Consider a directed graph with m nodes, $m \geq 2$.
- ▶ Consider an origin node o and a destination node d .
- ▶ There is a finite number of simple paths between o and d .

Finite number of simple paths

Lemma 21.6

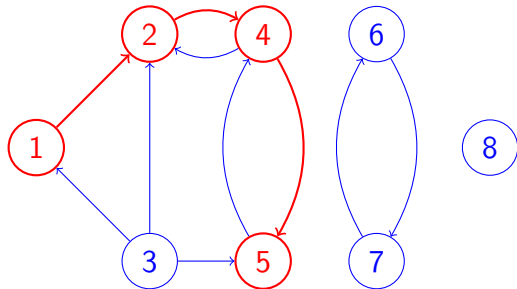
- ▶ Consider a directed graph with m nodes, $m \geq 2$.
- ▶ Consider an origin node o and a destination node d .
- ▶ There is a finite number of simple paths between o and d .

Proof

- ▶ Consider $2 \leq k \leq m$.
- ▶ Each simple path containing k nodes corresponds to a permutation of $k - 2$ nodes.
- ▶ For each k , the number of permutations is finite.
- ▶ As $k \leq m$, the total number is finite.

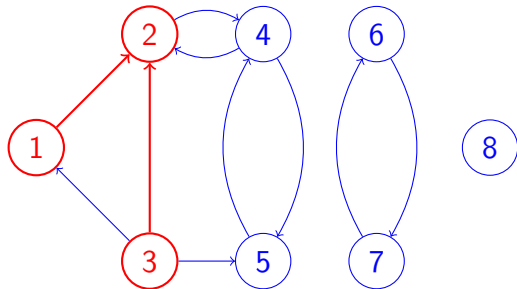
Connectivity

Nodes 1 and 5 are strongly connected.



Connectivity

Nodes 1 and 3 are connected.



Equivalence

Reflexive: a node is supposed to be connected to itself.

Symmetric: Mention that the relation “is strongly connected with” is not symmetric, and does not represent an equivalence class.

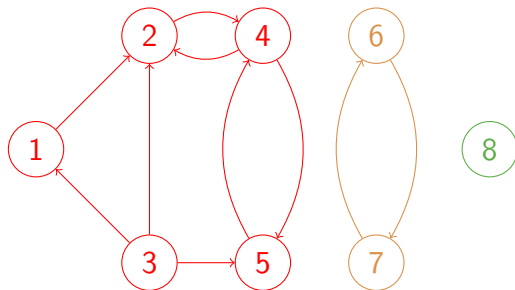
Equivalence class

“is connected with”

- ▶ reflexive,
- ▶ symmetric,
- ▶ transitive.

Connected component

- ▶ subgraph $G' = (\mathcal{N}', \mathcal{A}', \phi')$,
- ▶ $\mathcal{N}'$ is an equivalence class on $\mathcal{N}$ for the relation “is connected with”.



Summary

Simple paths have a finite length, and are finitely many.

The presence of a path between two nodes connect them.

Connected components are groups of nodes that are mutually connected

A connected graph has exactly one connected component.

All nodes are mutually connected.